Highlights

Field accuracy is not heat accuracy: wall-flux-constrained surrogates for non-Boussinesq convection

- 0.09% field error can coexist with 2.0% integral Nu error (200 samples).
- Random-wall replication: 0.47% field error, 4.04% Nu error (22 samples).
- Wall-flux penalty cuts Nu error to 0.01% and 0.58% (six-point).
- Boussinesq–compressible gap factorises as $D = \varepsilon P(X)$ over 627 pairs.
- External ε factorisation transfers within 0.2% without retraining.